

The empirical recurrence for OEIS A299749, recovered from its tail

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Abstract

OEIS A299749 records “Empirical recurrence of order 65” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 67$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 67$. Its last coefficient is $c_{65} = 172032 \neq 0$, so no further tail satisfies anything shorter; any order-65 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A299749 is “Number of nX4 0..1 arrays with every element equal to 0, 1, 2, 3, 5, 6 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

8, 88, 375, 1998, 11733, 64703, 358219, 2010841, . . .

The entry states:

Empirical recurrence of order 65 (see link above)

As of the “Last modified” line on the live entry (Feb 18 2018 10:30:41, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 67$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 67$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 65: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 65.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 134$ exact terms from $n = 2$ on, it returns order 65 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{65} c_i a(n-i),$$

c_1	10	c_{18}	1645178	c_{35}	−2212602232	c_{52}	172504786
c_2	−35	c_{19}	−7049773	c_{36}	1531580061	c_{53}	−244603126
c_3	128	c_{20}	20188236	c_{37}	−532405022	c_{54}	176055478
c_4	−609	c_{21}	−37764344	c_{38}	−732607513	c_{55}	77117224
c_5	1648	c_{22}	64941317	c_{39}	1367236914	c_{56}	−151621252
c_6	−4158	c_{23}	−114709662	c_{40}	−1555971813	c_{57}	99256984
c_7	13776	c_{24}	173518181	c_{41}	2536278488	c_{58}	−40251504
c_8	−29946	c_{25}	−259793291	c_{42}	−3675759109	c_{59}	−4412488
c_9	60069	c_{26}	385861213	c_{43}	2855831413	c_{60}	17600936
c_{10}	−150657	c_{27}	−519313686	c_{44}	−1236897149	c_{61}	−8123744
c_{11}	264220	c_{28}	743424815	c_{45}	198148955	c_{62}	1058752
c_{12}	−439913	c_{29}	−992057743	c_{46}	731880141	c_{63}	−126016
c_{13}	821566	c_{30}	1247226648	c_{47}	−1314833256	c_{64}	−217600
c_{14}	−1082187	c_{31}	−1610339578	c_{48}	1141038437	c_{65}	172032
c_{15}	1447892	c_{32}	1874418465	c_{49}	−219804872		
c_{16}	−1590746	c_{33}	−2003747304	c_{50}	197065284		
c_{17}	226776	c_{34}	2239254790	c_{51}	−369341660		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 67$, and it is the recurrence OEIS A299749 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{65} - c_1 x^{64} - \dots - c_{65}$. Its constant term is $-c_{65} = -172032$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and

the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 65 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 65, so $\deg q = 65$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 65, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 65, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 172032.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-65 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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